Session 06 Phase 3 – Master Results Tables

August 4, 2026

What each condition means

Every row below is a different way of turning the model's internal activity while it writes an answer into a single feature vector. Think of that internal activity as a trail of numbers – one per layer, per generated token – and each condition as a different way of boiling that trail down.

core_max The baseline. Takes the single highest value each internal signal ever reaches while generating the answer – a snapshot of the loudest moment.

q_velocity Instead of the activity itself, tracks how much it *changes* from one layer to the next – speed instead of position. Picks up on whether the model's internal representation is still being revised as information moves through the network, not just where it ends up.

q_static The same kind of peak snapshot as core_max, but softer: uses a high/low percentile instead of the literal maximum/minimum, so one unusual token can't skew the whole signal.

core_concat Staples core_max's peak snapshot and q_velocity's change signal together into one longer feature vector – "where it ended up" plus "how it got there."

joint_tensor Instead of computing two signals separately and combining them afterward, fits one shared compression across the combined raw data from the start, so the signals can influence each other during compression rather than only being stapled together at the end.

triple_concat Staples all three single signals – core_max, q_static, and q_velocity – together. The "use everything we have" condition.

Model	Condition	TruthfulQA	TriviaQA	NQ-Open	TyDiQA-GP
LLaMA-3.1-8B	HARP (published, supervised)	88.5	92.9	89.4	86.6
	HARP (our reimplementation)[§]	*pending*	*pending*	*pending*	*pending*
	core_max	84.4	86.4	85.7	83.1
	q_velocity	**86.2**	85.8	84.6	82.6
	q_static	84.3	86.7	86.1	**84.0**
	core_concat	86.0	86.8[†]	85.3	83.2
	joint_tensor	84.5	86.4	**86.3**	83.3
	triple_concat	85.5	**87.0**	85.7	83.2
Qwen2.5-7B	HARP (published, supervised)	88.1	92.8	84.0	88.4
	HARP (our reimplementation)[§]	*pending*	*pending*	*pending*	*pending*
	core_max	87.1	91.3[†]	78.8	**88.3**[‡]
	q_velocity	87.7	90.9	78.2	87.0
	q_static	87.1	91.7	79.1	87.5
	core_concat	87.8	91.5	**79.4**	87.8
	joint_tensor	87.2	**91.8**	78.5	87.0
	triple_concat	**87.8**	91.7	79.3	87.9

Table 1: **Literature-comparable results: pooled AUROC (%) under HARP's evaluation protocol** (train on 75% of known questions; test on the rest plus all fully-hallucinated questions), 5-seed mean, Random Forest readout. All non-gray rows are *our* unsupervised feature variants scored under this protocol; the gray row is the published score of HARP's own supervised detector, copied from their paper for reference (not recomputed by us). Bold = best of our conditions per column (HARP row excluded). [†]On TriviaQA – and *only* on TriviaQA – logistic regression outperforms the forest, for *both* models, which makes this a property of the dataset rather than the architecture. LLaMA: core_max 87.3 ± 0.2, q_static 87.3 ± 0.3, core_concat 87.3 ± 0.2. Qwen: q_static 92.7 ± 0.2, core_concat 92.6 ± 0.2, joint_tensor 92.5 ± 0.1, triple_concat 92.5 ± 0.1, core_max 92.4 ± 0.1 – LR wins all six conditions, and q_static sits 0.11 points below HARP's published 92.8, our closest approach to the published reference anywhere in the study. [‡]Qwen/TyDiQA-GP is the first cell where an unsupervised condition of ours reaches the published supervised score: core_max 88.3 ± 1.8 against HARP's reported 88.4, a gap of 0.1 points, well inside our seed-to-seed spread.

[§]**An important caveat on every comparison in this table.** We replicate HARP's evaluation *protocol* – their 75/25 known split, 10 beams per question, and their exact labelling rule (ROUGE-L > 0.7 or semantic similarity > 0.5) – but not their *method.* The published row was computed by them on *their* generations; our rows are computed on ours. Same protocol, different samples. Until we run their reasoning-subspace projection on our own pinned generations, "we match HARP" means "our number on our data is close to their number on their data," which is weaker than a like-for-like result. The reimplementation row above is reserved for that comparison.

Model	Condition	TruthfulQA	TriviaQA	NQ-Open	TyDiQA-GP
LLaMA-3.1-8B	core_max	81.77	81.97	76.10	84.48
	q_velocity	83.58	81.17	75.44	85.68
	q_static	83.28	81.97	76.72	85.26
	core_concat	**84.35***	82.06	**76.74**	85.27
	joint_tensor	83.76	81.84	75.70	85.13
	triple_concat	83.28	**82.27**	76.01	**85.76**
Qwen2.5-7B	core_max	82.47	88.56	74.45	87.39
	q_velocity	84.33	88.11	74.88	88.50
	q_static	83.28	88.70	75.32	86.88
	core_concat	**84.43**	88.55	76.18	88.85
	joint_tensor	82.69	**88.80**	75.87	87.11
	triple_concat	84.05	88.64	**77.61**	**89.14**

Table 2: **Within-prompt AUROC (%), grouped 5-fold CV, Random Forest.** Among the 10 answers to the *same* question, does the detector score the hallucinated ones above the truthful ones? Formally: concordant (hallucinated, truthful) pairs over total such pairs, counting only pairs drawn from the same question, ties at one half. No HARP row appears because no prior work in this area reports a per-question metric – see the preceding section for why we introduce one and what it costs. Bold = best per column, but note that a bolded lead is *nominal* unless marked *: on TriviaQA, NQ-Open, and TyDiQA-GP no condition's within-prompt gain over core_max survives a paired bootstrap (all 95% CIs include zero; triple_concat's TriviaQA lead, for instance, is $\Delta = +0.30$ pts, CI $[-0.14, +0.73]$). *core_concat vs. core_max on TruthfulQA remains, across all four completed datasets, the project's *only* statistically significant within-prompt improvement (paired $\Delta = +2.57$ pts, 95% CI $[+0.03, +5.26]$, excludes zero). Separately, all four combination conditions *do* significantly beat q_velocity on TriviaQA on both metrics. Because the leads are nominal, the practically useful reading of this table is the *level* each model reaches, not which condition tops a column.

Model	Condition	TruthfulQA	TriviaQA	NQ-Open	TyDiQA-GP
LLaMA-3.1-8B	core_max	+5.15	+3.44	+10.34	−2.39
	q_velocity	+4.87	+3.53	+10.96	−3.52
	q_static	+4.22	+3.67	+10.00	−2.31
	core_concat	+4.22	+3.54	+10.18	−2.58
	joint_tensor	+4.42	+3.61	+10.81	−2.44
	triple_concat	+5.25	+3.55	+10.72	−3.08
	mean	**+4.69**	**+3.56**	**+10.50**	**−2.72**
Qwen2.5-7B	core_max	+4.84	+2.40	+13.02	+0.13
	q_velocity	+4.36	+2.65	+13.43	−0.42
	q_static	+4.42	+2.58	+13.19	+1.31
	core_concat	+4.17	+2.61	+12.71	−0.45
	joint_tensor	+5.47	+2.62	+12.52	+0.77
	triple_concat	+4.51	+2.67	+11.10	−0.74
	mean	**+4.63**	**+2.59**	**+12.66**	**+0.10**

Table 3: **The difficulty gap: pooled minus within-prompt AUROC (percentage points).** Both terms come from the *same* grouped 5-fold run and the same Random Forest scores, so the difference isolates exactly one thing – how much a condition's pooled number depends on being allowed to compare answers across different questions. Positive means pooled *overstates* per-response detection; negative means it understates it.

Colour key. Shading is a diverging scale on the signed gap: warm ( ) where pooled overstates, cool ( ) where it understates, with intensity proportional to magnitude on a common scale across the whole table (deepest shade = 13.4 points, the largest gap observed). Near-white cells are gaps close to zero, where the two metrics agree.

The striking pattern is how little the gap moves down a column and how much it moves across one. Within any model×dataset cell the six conditions span about one point; across datasets the means run from −2.7 to +12.7. **The gap is a property of the benchmark, not of the feature variant.** NQ-Open is the extreme – at 89–97% hallucination rates it has very few mixed questions, so almost all of its pooled comparisons are cross-question and its pooled score is inflated by 10–13 points. TyDiQA-GP is where the two metrics most nearly agree: slightly negative for LLaMA (−2.7) and essentially zero for Qwen (+0.1). It also carries the smallest question-difficulty share in Figure 3 (66%, against 74–85% elsewhere), which is consistent with a near-zero gap. We report the LLaMA inversion without claiming a mechanism – the same dataset behaves differently under the two models, which argues against a purely structural explanation. (Note: we use the *English-only* subset of TyDiQA-GP, 440 of 5,077 rows, so no cross-language effect is involved.)

Practical consequence: a pooled AUROC cannot be interpreted without knowing its benchmark's gap. Two published numbers of 86.5 mean different things if one comes from NQ-Open and the other from TyDiQA-GP.

Note: Figure 3's right panel reports the same quantity for LLaMA and differs by 0.1–0.5 points (+4.9 / +3.4 / +10.8 / −2.5 against this table's core_max row). Both are computed from the same features and folds; the small discrepancy is an implementation difference between the two code paths and is under investigation. This table is built directly from the Phase-3 result files, consistent with Tables 1 and 2.

Model	Condition	TruthfulQA		TriviaQA		NQ-Open		TyDiQA-GP	
		RF	LR	RF	LR	RF	LR	RF	LR
LLaMA-3.1-8B	core_max	✓			✓	✓		✓	
	q_velocity	✓		✓		✓		✓	
	q_static	✓			✓	✓		✓	
	core_concat	✓			✓	✓		✓	
	joint_tensor	✓			✓	✓		✓	
	triple_concat	✓			✓	✓		✓	
Qwen2.5-7B	core_max	✓			✓	✓		✓	
	q_velocity	✓			✓	✓		✓	
	q_static	✓			✓	✓		✓	
	core_concat	✓			✓	✓		✓	
	joint_tensor	✓			✓	✓		✓	
	triple_concat	✓			✓	✓		✓	

Table 4: **Which classifier wins, under the HARP protocol** – ✓marks the higher of RF/LR (5-seed mean pooled AUROC) for that condition×dataset cell. RF sweeps TruthfulQA, NQ-Open, and TyDiQA-GP (18/18 LLaMA cells). TriviaQA flips: LR wins 5 of 6 conditions, RF holds only on q_velocity. See Table 1's caption for the actual TriviaQA numbers (LR's wins there are narrow, 0.2–0.9pt – not a landslide like RF's wins elsewhere). Qwen reproduces this exactly: RF sweeps TruthfulQA, NQ-Open and TyDiQA-GP (18/18 cells) and LR sweeps all six TriviaQA conditions. Across both models, 42 of 48 condition×dataset cells favour RF and all 12 exceptions are TriviaQA. The inversion therefore tracks the *dataset*, not the architecture.

Within-prompt AUROC: motivation, definition, and why we introduce it

The convention, and what it silently assumes

Hallucination detection is evaluated almost universally with a single dataset-level AUROC. HARP states that "AUROC (area under the ROC curve) is employed as the primary evaluation metric"; INSIDE that "AUROC is a popular metric to evaluate the quality of a binary classifier"; and Janiak et al. that AUROC "assesses the ability of a hallucination detection method to correctly rank positive and negative instances (hallucinations vs. non-hallucinations)."

All three generate *multiple answers per question* – ten, in every case, as do we. So the data has an explicit two-level structure: answers nested inside questions. Yet none of the three states how scores from *different* questions are combined, and the standard implementation simply flattens the array. Every hallucinated answer is ranked against every truthful answer in the dataset, regardless of whether they answer the same question. The grouping is present in the data and discarded without comment.

Why that matters: the shortcut

Questions differ enormously in difficulty. In our data, 300 of TruthfulQA's 817 questions are ones where the model got all ten attempts wrong, and 111 where it got all ten right. Under pooled AUROC, every answer from the first group is compared against every answer from the second, and a detector gets all of those pairs correct simply by outputting a higher score for hard questions. It never has to distinguish a good answer from a bad one.

This is not hypothetical. Janiak et al. report two "simple baselines" – `Mean-Len` (mean answer length across a question's ten generations) and `Std-Len` (their standard deviation) – that are *per-question constants*: identical for all ten answers to a question, carrying literally zero information about any individual answer. On LLaMA they average 0.721 and 0.751 AUROC respectively, against 0.747 for Eigenscore, a sophisticated internal-state detector. A feature that *cannot* discriminate between two answers to the same question, even in principle, matches or beats the state of the art under pooled AUROC. They present this as evidence that simple baselines are competitive; we read it as evidence that the metric is measuring something other than what it appears to.

Definition

Both metrics are computed from the *same* score vector – one detector, trained once, scoring every answer out-of-fold. They differ only in which comparisons are permitted.

AUROC in either form is a pair-counting quantity: among all (hallucinated, truthful) pairs, the fraction in which the hallucinated answer received the higher score, with ties counting one half.

- **Pooled AUROC** forms every such pair in the dataset, including pairs drawn from different questions.
- **Within-prompt AUROC** forms only pairs whose two members answer the *same* question, then aggregates concordant pairs over total pairs across all questions.

A question whose ten answers are all truthful, or all hallucinated, yields no valid pair and drops out entirely. Only *mixed* questions – where the model both succeeded and failed on the same input – contribute.

"OOF" means out-of-fold. Under `GroupKFold(5)` grouped by prompt, all ten answers to a question stay together, so a question is never split across training and test. Each answer is scored by the one model that did not see it. "Pooled OOF AUROC" is a single AUROC over all held-out scores, not an average of five per-fold values.

Why this is the right correction, and not a novel one

Conditioning on a grouping variable to remove a confound is standard practice elsewhere. Information retrieval computes MAP and NDCG per query and then averages; pooling documents across queries would be considered plainly wrong. Recommender systems use per-user AUC ("GAUC") for exactly the reason we need it here: pooled AUC conflated "this user clicks often" with "this item suits this user." Substituting question for user and answer for item gives our setting exactly.

We therefore do not claim within-prompt AUROC as a new metric. We claim that hallucination detection has not been applying it, that the two-level structure required for it is already present in every standard benchmark, and that the difference it makes is large – which is what Figure 3 quantifies.

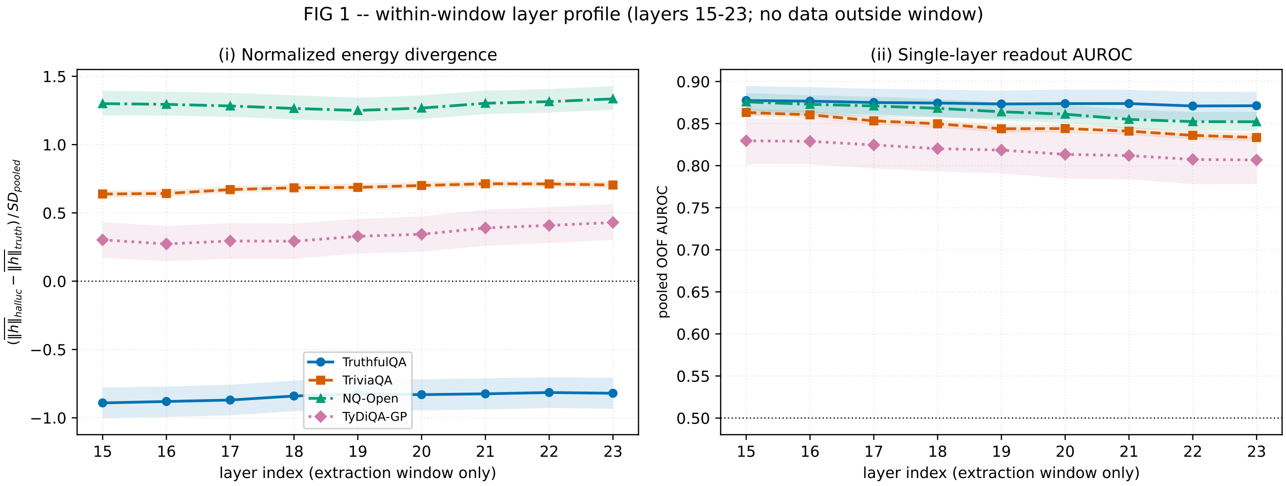
What it costs

The correction is not free, and the limitations should be stated plainly.

- **Homogeneous questions are discarded.** On Qwen/NQ-Open only 348 of 3,610 questions are mixed, so the metric rests on 5,365 pairs rather than the full dataset, and its confidence intervals are correspondingly wide. Where the model almost never succeeds, there is little to measure.
- **It is not comparable to published numbers.** No prior work reports it, so pooled AUROC must still be reported alongside for literature comparison. We report both throughout.
- **It is not uniformly stricter.** On TyDiQA-GP under LLaMA, within-prompt AUROC *exceeds* pooled by 2.7 points – restricting comparisons to the same question helps there rather than hurting. The same dataset under Qwen sits at essentially zero. The sign of the gap is a per-benchmark diagnostic, not a fixed penalty, and it cannot be assumed in advance.

The gap between the two numbers is the quantity of interest, and Figure 3 decomposes it directly.

Figure 1 – where in the network the signal lives

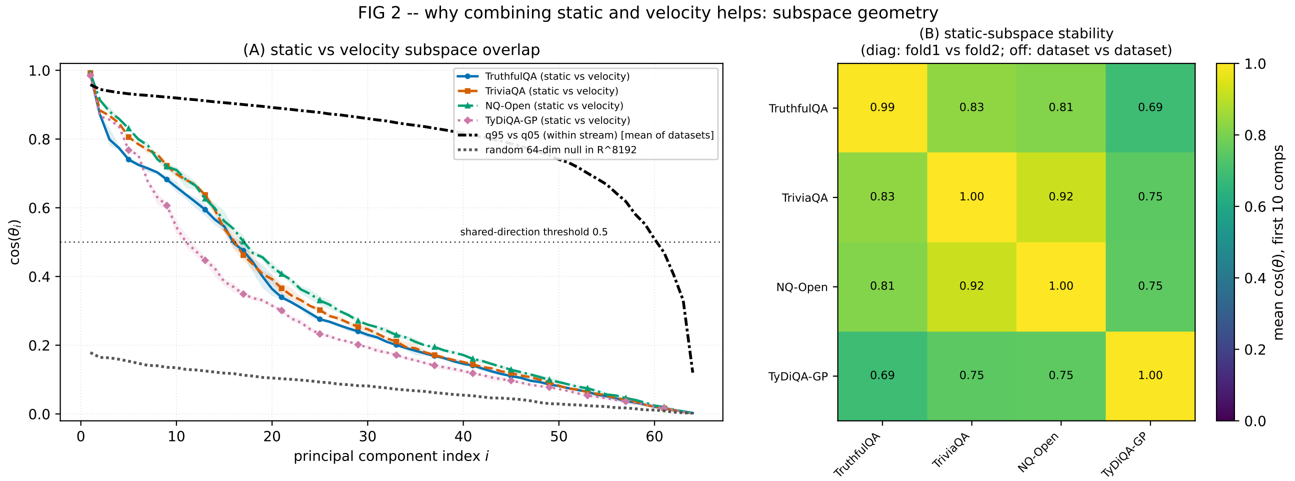


LLaMA-3.1-8B. Left: how much larger the average size of the internal activity is for hallucinated answers than for truthful ones, in pooled standard deviations. Right: detection score when the detector is allowed to use only that one layer (64 components, logistic regression). Bands are 95% confidence intervals. Only layers 15–23 exist on disk.

What it shows

- **Left panel: the curves are flat.** There is no bump at any layer. The "layer-22 hump" we have been assuming does not exist – the whole range of movement across nine layers is about 0.1 standard deviations, which is nothing.
- **The datasets sit far apart, and one has the opposite sign.** NQ-Open is at +1.3, TriviaQA +0.7, TyDiQA-GP +0.35, and TruthfulQA at −0.85 – there, hallucinated answers have *smaller* internal activity, not larger. So this quantity behaves like a fingerprint of the dataset, not a signal of hallucination.
- **Critically: this quantity does not predict detection at all.** NQ-Open has the largest gap and TruthfulQA has a negative one, yet both sit at the top of the right panel. Since the 15–23 window was originally chosen using exactly this kind of energy audit, the method we used to pick our window does not track what we actually care about.
- **Right panel: layer 15 is the best single layer on all four datasets.** Three of the four decline steadily to the right; TruthfulQA is flat rather than falling.
- **Layer 15 is the left edge of our window, which is a warning sign.** When the best value sits on the boundary of the range you looked at, the true best is usually outside it. The honest reading is that we may have been extracting from the wrong part of the network, and we cannot tell because the pipeline crops at 15. Settling this needs one GPU pass over earlier layers.
- **A single layer appears to match the entire pipeline.** Layer 15 alone, with 64 numbers and a simple linear model, scores about 0.877 / 0.862 / 0.873 / 0.830 across the four datasets, against 0.869 / 0.854 / 0.865 / 0.821 for the full 9-layer, 320-number random forest. The confidence intervals overlap, so the claim is "matches," not "beats" – but even a match means the other eight layers are contributing close to nothing. This needs one cheap paired check before we state it publicly.

Figure 2 – do the static and velocity streams carry different things?

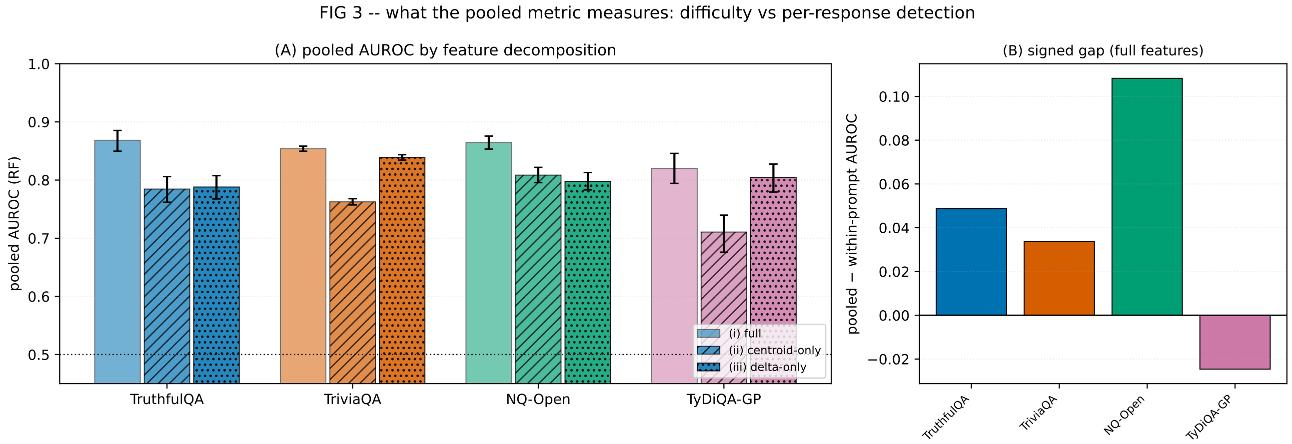


Left: for each pair of directions (1st most important, 2nd, and so on up to 64th), how closely aligned the static and velocity subspaces are. 1.0 means the two directions are identical; 0 means completely unrelated. Right: how stable the static subspace is, comparing folds within a dataset (diagonal) and datasets against each other (off-diagonal).

What it shows

- **Three clearly separated bands in the left panel.** The black line (the two halves of the *same* stream, q95 vs q05) stays near 1.0 almost the whole way – they are near-copies, sharing about 58 of 64 directions. The coloured lines (static vs velocity) drop below 0.5 around the 15th direction – about 15 of 64 shared. The grey line (two random subspaces) never rises above 0.18 – nothing shared.
- **So static and velocity really are geometrically different** – far more different than chance, far less different than two halves of one stream. That part of the story holds.
- **But being different did not make them useful together.** Combining the two streams buys only 0.1–2.5 points, mostly not statistically significant. So the two streams point in different directions while carrying largely the *same* information about hallucination. This is the honest headline, and it explains why several rounds of feature engineering produced about one point.
- **Where the overlap sits matters.** The first five directions are nearly identical (above 0.8). The part the two streams share is their biggest, most dominant part – which is plausibly the part a classifier leans on most. The genuinely different directions are the small, tail ones.
- **A free simplification.** Because q05 is nearly a copy of q95, we could likely drop q05 entirely and halve the feature size at no cost. That is a CPU-only check on data we already have.
- **Right panel, and a good rigour point:** comparing two different folds of the same dataset gives 0.99–1.00. Our unsupervised basis is essentially deterministic – it does not wobble when the data is resampled. Across different datasets it is 0.69–0.92, so there is substantial shared structure, with TyDiQA-GP the most distinct.
- **One caution when presenting:** the diagonal and off-diagonal cells of that heatmap measure two different comparisons. It is stated in the title but is easy to miss.

Figure 3 – what the standard metric is actually measuring



Left: pooled AUROC using full features, using only each question's 10-answer average ("centroid-only," which destroys all per-answer information), and using each answer minus that average ("delta-only," which destroys all question-identity information). Right: pooled AUROC minus within-prompt AUROC, per dataset.

What it shows

- **Question identity alone gets most of the way there.** The centroid-only bars reach 0.71–0.81. Those features contain *zero* information about any individual answer – every answer to a question has been replaced by the same average. Yet they recover 66–85% of the pooled score's lift above chance.

- **What this does and does not mean.** It does not mean the metric is invalid – those features still come from the model's own internal states. It means pooled AUROC is mostly measuring *which questions the model will fail on*, not *which specific answers are lies.* Most readers of a hallucination-detection paper assume the second.

- **The size of the distortion varies a lot by benchmark** (right panel). NQ-Open is the most inflated at +0.108, then TruthfulQA +0.049 and TriviaQA +0.034. TyDiQA-GP is the only one where pooled *understates* performance, at −0.025.

- **The most useful result is not currently drawn in the figure.** Removing the question average and keeping only the per-answer part *beats* the full features on within-prompt AUROC: 83.2 vs 82.0 on TruthfulQA, 83.8 vs 82.0 on TriviaQA. So the question average is not just unhelpful for per-answer detection – it is actively getting in the way.

- **That is a plain subtraction beating all six engineered feature variants** on the metric we care about, at no compute cost. It should go into the condition grid for both models.

- **One honest caveat on it.** Subtracting the average needs the other nine answers available at test time, so it is a "multi-sample" method – the same setting as SelfCheckGPT and INSIDE, which is legitimate and precedented, but it is not a like-for-like comparison against single-pass HARP and must be labelled as such. It uses no labels, so there is no leakage.

- **This lines up with the tables.** Combination conditions beat the baseline on pooled but never on within-prompt, for both models and every dataset – exactly what you would expect if those combinations are adding question-difficulty information rather than per-answer signal. Two separate analyses reaching the same conclusion.

Novel contributions

Claims are graded by how well the current data supports them. [**Established**] means our own measurements support it directly. [**Needs lit check**] means the result is solid but we have not confirmed no one has published it. [**Do not claim**] records things we deliberately are *not* asserting, so the boundary is explicit.

Part 1 – What we have now

1. Pooled AUROC is mostly a question-difficulty detector, and we measured how mostly. [**Established, needs lit check**] Replacing every answer by its question's 10-answer mean destroys all per-answer information, yet those features recover **66–85% of pooled AUROC's lift above chance** across four datasets. The centroid/delta decomposition is the instrument, and it is the contribution – not the observation that a confound exists, but a number for how large it is. Janiak et al. independently produced supporting evidence without framing it this way: their `Std-Len` baseline, a per-question constant that cannot separate two answers to the same question even in principle, scores 0.751 pooled AUROC against Eigenscore's 0.747.

2. The difficulty gap is a property of the benchmark, not of the method. [**Established**] Across 48 model×condition×dataset cells, the pooled-minus- within-prompt gap varies by about one point across the six conditions within a cell, and by *fifteen* points across datasets (-2.7 to $+12.7$; Table 3). The practical consequence is concrete and checkable: two published pooled AUROCs of 86.5 describe different things if one comes from NQ-Open and the other from TyDiQA-GP. This motivates reporting the gap as a benchmark descriptor – see Part 2.

3. The sign of the gap varies, so this is a diagnostic and not a correction factor. [**Established; mechanism unexplained**] TyDiQA-GP inverts for LLaMA: pooled *understates* per-response detection there by 2.7 points, while the same dataset under Qwen sits at essentially zero. Any framing that treats pooled AUROC as uniformly inflated is wrong. We report the inversion as an observation and do not offer a mechanism; that the two models disagree on the same data argues against a purely structural explanation, and identifying the cause is an open question.

4. Energy divergence does not track discriminability, which invalidates a common layer-selection heuristic. [**Established, needs lit check**] Figure 1: the norm gap between hallucinated and truthful answers is essentially flat across the extraction window ($\sim$0.1 SD over nine layers), varies enormously between datasets ($+1.3$ to -0.85), *flips sign* on TruthfulQA, and is uncorrelated with per-layer detection AUROC – NQ-Open has the largest norm gap and TruthfulQA a negative one, yet both rank highest on discriminability. Our own extraction window was selected by exactly such an audit, so this is a self-critique with general force.

5. Geometric complementarity does not imply informational complementarity. [**Established**] Figure 2: the static and velocity subspaces share only $\sim$16 of 64 directions against a random null of 0 and a within-stream reference of $\sim$58 – genuinely different geometry – yet combining them buys 0.1–2.5 points, mostly not significant. The overlap concentrates in the leading, highest-variance directions, which is plausibly what a classifier leans on. This explains why several rounds of feature engineering produced about one point, and it is a more useful negative result than "the combination did not help."

6. A plain mean-subtraction outperforms all six engineered feature variants. [**Established on 2 of 4 datasets; extension pending**] Removing each answer's prompt centroid beats the full features on within-prompt AUROC (83.2 vs 82.0 on TruthfulQA, 83.8 vs 82.0 on TriviaQA) at zero compute cost. Must be reported as a *multi-sample* method, since centering requires the sibling answers at inference – the SelfCheckGPT/INSIDE setting, legitimate and precedented, but not comparable to single-pass HARP.

7. An unsupervised pipeline reaches published supervised performance. [Established] On Qwen2.5-7B, three of four datasets sit within 0.3 points of HARP's published supervised scores: TyDiQA-GP −0.1, TriviaQA −0.11, TruthfulQA −0.3. The best condition on two of them is plain `core_max`, which supports the reading that the *pipeline* is the contribution rather than any particular feature variant.

8. Two smaller consistency results. [Established] The RF/LR inversion tracks the dataset, not the architecture: 42 of 48 cells favour Random Forest and all 12 exceptions are TriviaQA, for both models. And condition ranking is stable across architectures with different hidden dimensions (4096 vs 3584) – `core_concat`/`triple_concat` occupy the top two positions in 5 of 6 completed cells.

What we explicitly do not claim. [Do not claim] Within-prompt AUROC is *not* a new metric – information retrieval computes MAP and NDCG per query, and recommender systems use per-user AUC (GAUC) for precisely this confound. Our claim is that hallucination detection has not been applying it despite every standard benchmark already having the required structure. Nor do we claim to have discovered that hallucination benchmarks are confounded; Janiak et al. established that for labels and response length. Ours is a different axis.

Part 2 – Future contributions reachable from the current state

Ordered by expected value per unit cost. All but the last two are CPU-only on artifacts already on disk.

A. Re-label with an LLM judge and re-run everything. [GPU-free, API cost. Highest value.] Janiak et al. show ROUGE-based correctness labels have 0.401 precision against human judgment, and that detector rankings change substantially – up to 45.9% – when labels are corrected. *Our labels are ROUGE-L > 0.7 or BLEURT > 0.5, so we inherit this criticism in full.* Every generation is already pinned to disk and labelling is a separate pipeline stage, so re-labelling and re-running costs no GPU time. This is simultaneously our largest exposure and the cheapest way to close it: if our findings survive human-aligned labels, they become far more defensible, and "we re-verified under LLM-as-Judge" is itself a contribution.

A2. Reimplement HARP and run it on our own generations. [Cost unscoped – see below.] Every comparison in Table 1 is currently our method on our data against HARP's published number on their data. The protocol matches theirs exactly, but the generations do not, so "we tie HARP" is not yet a like-for-like claim and will be the first thing a reviewer asks about. Running their reasoning-subspace projection on our pinned generations fixes that.

It is worth more than a fairness repair, for two reasons. First, it *neutralises the labelling criticism for the comparison specifically*: if both methods are scored against the same labels, then even if those labels are imperfect (contribution A), the *relative* result stays valid. Second, and more valuable, it lets us compute HARP's *within-prompt* AUROC and its difficulty gap. At present our decomposition shows that *our* features are 66–85% question difficulty. Showing the same for a published state-of-the-art detector would lift that finding from a property of our pipeline to a property of the field's methods – which is a substantially stronger claim than anything else in Part 1.

Cost is currently unscoped. We have read HARP's evaluation protocol but not their method section in detail, so we do not yet know whether their projection can be computed from the pooled tensors already on disk or requires a fresh extraction pass. That determination should precede any commitment.

B. Does one layer match the whole pipeline? [CPU, hours.] Figure 1(ii) suggests layer 15 alone, at 64 dimensions with logistic regression, matches the full 9-layer 320-dimension random forest on all four datasets (0.877/0.862/0.873/0.830 vs 0.869/0.854/0.865/0.821). Confidence intervals overlap, so the verifiable claim is "matches," not "beats" – but a match means eight of nine layers contribute nothing and the compression story strengthens fivefold. Requires one paired test against `core_max` on identical folds.

C. Is the extraction window in the wrong place? [One GPU pass. Highest ceiling.] The best single layer is 15 – the *left edge* of the window. An optimum sitting on a boundary usually means the true optimum lies outside it. If the signal peaks before layer 15, then the convention of extracting from middle-to-late layers may be misplaced across this whole line of work, and every number we have is a lower bound. One extraction pass over layers ∼7–15 on two datasets settles it. Note the caveat: single-layer AUROC is not marginal contribution inside a 9-layer fit, so a monotone profile is suggestive rather than conclusive.

D. Promote delta-centering from diagnostic to method. [CPU, free.] Extend contribution 6 across the full grid and both models. If it holds, the paper gains a concrete improvement that follows directly from the decomposition – the diagnostic and the method share one idea, which is a much tighter story than reporting them separately.

E. Report the gap as a benchmark descriptor. [CPU, near-free.] Turn contribution 2 into a proposed practice: alongside any pooled AUROC, report that benchmark's difficulty gap so readers can interpret the number. We would publish gaps for the four standard benchmarks as a reference table. This is the cheapest path from "we found a problem" to "we gave the community a fix."

F. Aggregate properly instead of testing cell by cell. [CPU, hours.] Combination conditions beat the baseline on pooled but never on within-prompt in *any single cell*. A random-effects meta-analysis across all eight cells, plus a Friedman/Nemenyi test over the six conditions, resolves whether that is a real null or eight underpowered tests. Either outcome is publishable, and "this subfield tests underpowered comparisons one dataset at a time" is a methods point in its own right.

G. Halve the feature dimensionality for free. [CPU, hours.] Figure 2 shows the q95 and q05 halves share ∼58 of 64 directions. If dropping q05 costs nothing measurable, every feature variant halves in size. Minor alone; useful combined with B.

H. Cross-dataset transfer. [CPU, heavy.] Fit on one dataset, evaluate on another, all 16 pairs per model, against HARP's published transfer matrix. Tests whether the learned subspace is general or dataset-specific – and Figure 2(B) already gives a geometric prediction to check it against (cross-dataset subspace similarity 0.69–0.92, TyDiQA-GP most distinct), so the two analyses can corroborate each other.

Sequencing. A and A2 are the two validity items and should lead. A determines whether our labels are trustworthy at all; A2 determines whether our headline comparison is like-for-like. They are complementary – A2 makes the *comparison* robust to label error even before A makes the *absolute numbers* trustworthy – so A2 is worth doing even if A slips. B, D, E and F are cheap, independent, and can run alongside. C has the highest ceiling but the highest cost, and is best attempted once A has confirmed the findings survive corrected labels. G and H are consolidation.